Extended Kalman Filter Fusion of Visual Detection and Odometry Under Sparse Observation: A Multi-Robot Digital Twin Case Study

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Abstract. In the digital twin studied here, multi-robot systems combining odometry with external visual detection relied on a discrete fallback: trusting detection when available, odometry otherwise. This approach discards the uncertainty of each estimate and offers no well-defined behavior during detection gaps. To address this, we replace the fallback with a per-robot Extended Kalman Filter fusing odometry and YOLO-based visual detection from a fixed overhead camera, validated in two stages: first statistical consistency on synthetic data (NEES/NIS), then evaluation on real recorded data, on a ROS2/Gazebo digital twin with three TurtleBot3 robots.
The result depends on how error is measured. At correction instants, the filter reduces position error by +23.7% relative to odometry alone under aggressive drift, the gain the literature typically reports. But measured continuously, at every odometry reading, the same filter is 12.7% *worse* than raw odometry, under the sparse visual coverage (0–28%) our camera exhibits in practice. We show this sign reversal is the expected consequence of operating below a critical observation rate, not a design flaw, and connect it mechanistically to a covariance ceiling required to keep Mahalanobis-gated association stable. Reporting both metrics, rather than only the commonly cited correction-instant gain, is necessary to honestly characterize EKF fusion under sparse observation; we also discuss why alternative filters (UKF, particle filter, IMM) do not resolve this limitation, since it stems from the observation rate, not the choice of filter.

Keywords: Extended Kalman Filter · Intermittent Observations · Cooperative Localization · Asynchronous Sensor Fusion · Multi-Robot Localization

1 Introduction

Fixed overhead cameras observing multiple mobile robots are common infrastructure in warehouse automation, indoor fleet coordination, and smart-environment monitoring, where installing a dedicated sensor payload on every robot is costly or impractical. Multi-robot systems combining onboard odometry with such external visual detection face a recurring choice: how to combine two noisy position estimates that arrive at different rates and are not always available at the same instant. A common, pragmatic solution, used in the digital twin studied here prior to our contribution, is a *discrete fallback*: trust detection when confidently associated, odometry otherwise. This approach discards information: no partial correction from low-confidence detections, no uncertainty propagation, no well-defined behavior during detection-free stretches.

Extended Kalman Filtering (EKF) is the classical answer to this problem, offering full probabilistic prediction and correction with explicit uncertainty propagation. Its convergence properties, however, are classically established under reasonably dense, regular observation [5], an assumption a single fixed overhead camera, subject to finite field of view and self-occlusion during maneuvers, does not generally satisfy. This paper investigates whether that violation matters in practice.

We replace the discrete fallback with a per-robot EKF fusing odometry and YOLO-based visual detection, evaluated end-to-end on the same digital twin. Our contribution is twofold. First, validated in two stages, synthetic NEES/NIS and then real recorded data, the filter delivers a real fusion gain: +23.7% position error reduction at correction instants, under aggressive drift. Second, the same filter, measured continuously and not only at correction instants, is 12.7% worse than raw odometry under the sparse visual coverage (0–28%) our camera exhibits in practice. We explain this reversal mechanistically, connecting it to established theory on Kalman filtering under intermittent observations [6,7]. We argue that reporting both metrics is necessary for honest characterization, and discuss why alternative filters do not resolve the underlying limitation.

2 Related Work

Durrant-Whyte and Bailey [4] formalize the standard EKF motion/observation model our filter follows directly (odometry drives prediction, YOLO detections drive correction; notation adopted in Section 3). Dissanayake et al. [5] prove EKF-SLAM covariance convergence under *dense* landmark observation, an assumption our 0–28%-coverage camera violates by construction (central to Section 4). Sinopoli et al. [6] establish that, under Bernoulli observation arrival, covariance diverges below a critical rate; our Scenario 3 (0% observed coverage) is consistent with the limiting case $p \to 0$, so the saturation we observe is theoretically expected, not a symptom of misconfiguration. Censi [7] notes this critical threshold depends on the covariance summary metric used; we report trace(P) as a scoping choice, not a claim of metric-independence.

Khosravi et al. [10] report decentralized cooperative localization for two robots (mutual LiDAR corrections), with 34%/56% RMSE reduction, comparable in order of magnitude to our +23.7%. Ahmed and Frémont [11] fuse a custom YOLOv8-RGBDis detector with a per-target EKF for a maritime multi-robot team (mAP@0.5 = 0.995, numerically close to ours). Neither reports a continuous-error regime in which fusion underperforms the unfused baseline: the differentiating finding of this work. Dobrynin and Garcia [12] share our exact platform (three TurtleBot3, ROS2) but solve camera-free relative localization, so do not encounter sparse visual coverage (revisited as complementary work in Section 5). Our contribution is the joint report of both the correction-instant gain and the continuous-error degradation, grounded in [6,7] for why this trade-off is expected rather than an implementation bug. We also do not evaluate against a public multi-robot benchmark such as UTIAS MRCLAM [13] or S3E [14]: both assume a sensor payload carried by each robot, whereas our setup observes robots from a single external, static overhead camera, closer to a fixed smart-environment sensor than to cooperative SLAM. To our knowledge no public dataset combines these three elements, which motivates the recorded dataset released with this work (Section 3). This coverage reflects targeted search, not a systematic literature review with a registered protocol; a formal survey remains open.

3 Method

Following Durrant-Whyte and Bailey [4], each robot is tracked by an independent EKF with state $x_k = [p_x, p_y, \theta]^\top$. The motion model integrates the *delta* of consecutive odometry readings onto the current state, not the absolute value, which would discard prior visual corrections. The observation model maps state to the 2D position reported by a YOLO detector on a fixed overhead camera (Section 3).

Prediction and covariance capping. Process noise is $Q = \mathrm{diag}(0.5, 0.5, 0.25)$, scaled by $\Delta t/0.1$ for the real odometry rate (≈29 Hz vs. the 0.1s assumed during synthetic calibration); calibrated empirically (a sweep from 10^{-6} to 1.0 showed monotonically increasing gain up to saturation at $Q \geq 0.5$). A hard ceiling $\mathrm{clip}(P, -\infty, 0.05)$ (element-wise) is applied after each prediction: without it, P grew unbounded during long uncorrected intervals ($\mathrm{trace}(P) \approx 85$ observed vs. ≈ 0.5 expected), breaking Mahalanobis-gated association by widening a robot's acceptance region enough to claim a neighbor's detection. Both Q and this ceiling were calibrated solely against the correction-instant gain in Table 1; the continuous-error degradation reported below is therefore an observed consequence of that choice, not a jointly optimized trade-off. This is an engineering response to the divergence Sinopoli et al. [6] characterize for linear systems below a critical observation rate; the analogous result for the EKF is only local, conditional on small enough noise and initial error [2], so we treat the connection as mechanistic motivation rather than a formal guarantee for our nonlinear filter. The ceiling trades strict optimality for bounded, predictable behavior under sparse observation.

Data association and correction. Each detection is assigned via Mahalanobis-gated nearest-neighbor: $d^2 = e^\top P_{xy}^{-1} e$ per (robot, detection) pair, accepted only if $d^2 \leq 9.21$ (99% confidence, χ^2, 2 DOF). This inherits a known limitation (Altendorfer and Wirkert [9]): an inflated covariance widens the accep-

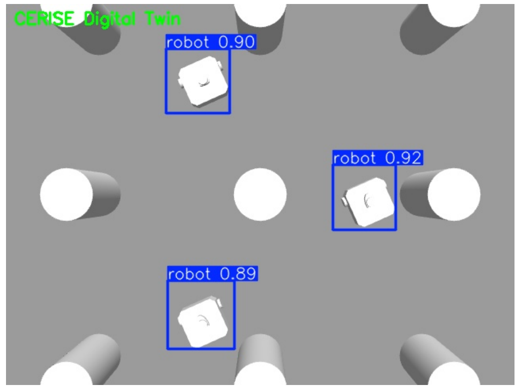

Fig. 1. Overhead view from the fixed camera over the three TurtleBot3 robots, with YOLO's bounding boxes and detection confidence overlaid in real time.

tance region and can bias association toward more uncertain tracks, the same failure mode motivating the covariance ceiling above. Once assigned, measurement noise R is derived linearly from YOLO detection confidence $c \in [0,1]$, $\sigma = R_{\min} + (1-c)(R_{\max} - R_{\min})$, $R_{\min} = 0.02$, $R_{\max} = 0.20\,\mathrm{m}$.

Camera calibration. Intrinsics come from `cv2.calibrateCamera` (Zhang [8]). A first attempt with all four intrinsics free reached low reprojection error (0.162px) but an ill-conditioned focal length (f_x 33% above the FOV-derived theoretical value): the degenerate case Zhang's method suffers when the checkerboard plane stays fronto-parallel to the sensor across all poses, unavoidable with our fixed downward-looking camera. We corrected this by fixing f_x, f_y, c_x, c_y at their geometric nominal values and calibrating only lens distortion, the standard remedy when nominal intrinsics are already known; the corrected calibration reproduces the FOV-derived heuristic already used in production to within 0.001m, confirming both are geometrically equivalent.

Validation. Following the standard consistency tests for Kalman filters [1], we first check NEES and NIS in simulation before trusting the filter on real data: 30 Monte Carlo seeds (2000 steps each) give NEES$= 2.955$ (expected 3.0) and NIS$= 1.982$ (expected 2.0), within 1.5%/0.9% relative error, well-calibrated despite formal χ^2 95% bounds being too narrow to pass at this sample size. The calibrated filter is then run against real odometry and YOLO detections recorded from the live digital twin, where observation coverage is sparse and irregular by construction, not by design choice.

Experimental setup. We evaluate our EKF in a ROS2/Gazebo digital twin: three TurtleBot3 Waffle units alongside a static overhead camera at $(0,0,3)\,\mathrm{m}$, $90°$ downward pitch, $60°$ horizontal FOV, 640×480@10 Hz. Each robot also publishes odometry ($\approx$29 Hz) and an unused 2D LiDAR scan (Section 5). Robot detection uses a YOLO model trained on 375 annotated frames from the same camera, mAP@0.5 $= 0.995$; confidence is carried in the detection message's otherwise-unused z field for direct consumption by the EKF correction step.

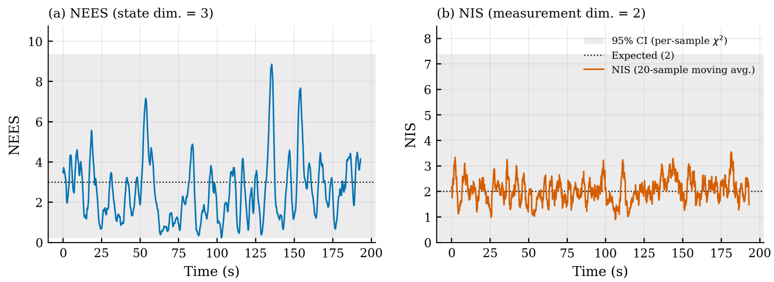

Fig. 2. NEES and NIS consistency over the synthetic Monte Carlo validation (30 seeds, 2000 steps each). Both statistics track their theoretical expected values, indicating a well-calibrated filter.

Fig. 1 shows the actual output of this detection pipeline: the three robots, as seen by the fixed overhead camera, with YOLO's bounding boxes and confidence overlaid.

Three ~18 s scenarios were recorded as ROS2/MCAP bags: (a) **Stationary**: 152 detections vs. 550 odometry readings/robot, ≈28% coverage, the highest of the three; (b) **Straight motion** ($v_x = 0.1\,\mathrm{m/s}$): 144 vs. 522 readings, ≈11% coverage; (c) **Curve/occlusion** ($v_x = 0.12\,\mathrm{m/s}$, $\omega_z = 0.4\,\mathrm{rad/s}$): 145 detections recorded, but **zero** successfully associated to `robot1` over the full recording, producing the 0% coverage case central to Section 4. Timestamp synchronization was verified via `rosbag2_py` (no gap >500 ms), ruling out sensor-clock desynchronization as a confound.

We release a reproducibility package (world file, URDF, calibration, EKF hyperparameters, code commit SHA) and document one non-obvious dependency: `ros-humble-rosbag2-storage-mcap` is not installed by default on ROS2 Humble.

4 Results and Discussion

We report two distinct error metrics, deliberately kept separate: *correction-instant error*, measured only when a YOLO detection is successfully associated and fused (most commonly reported in related work; Section 2), and *continuous error*, measured at every odometry reading regardless of correction. These two metrics disagree in sign, and both are necessary to honestly characterize the filter's behavior under sparse observation.

Correction-instant error. The synthetic NEES/NIS validation (Section 3) confirms the filter is well-calibrated before any real-data evaluation (Fig. 2, NEES→ 3.0, NIS→ 2.0).

Table 1 reports position error under three injected odometry drift regimes, aggregated across all three recorded scenarios ($n = 1323$). Under aggressive drift, where raw odometry has already accumulated substantial error before a visual correction becomes available, the EKF reduces mean position error by +23.7% relative to odometry alone (Wilcoxon signed-rank, $p < 0.001$; samples within

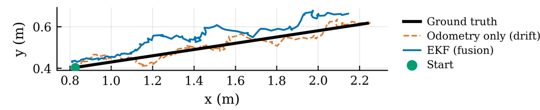

Fig. 3. Estimated trajectory under aggressive drift, Scenario 2 (straight motion). The EKF (blue) tracks ground truth (black) more closely than raw drifted odometry (orange, dashed).

a scenario are temporally autocorrelated, so this n overstates the number of independent trials, and the p-value should be read as optimistic, a limitation compounded by each scenario spanning only $\sim$18 s). Under light drift, the gain is negative (-5.3%): visual detection error ($\approx$3–6 cm) is not yet smaller than the still-small odometry error, so fusion offers no benefit, expected behavior for a correctly calibrated filter, not a failure.

Table 1. Correction-instant position error (m), aggregated over 3 scenarios ($n = 1323$). Each cell reports EKF / odom-only; Δ columns give EKF's relative reduction.

Drift	**Mean EKF/Odom**	**Mean Δ**	**RMSE EKF/Odom**	**RMSE Δ**
None (odom=GT)	0.036 / 0.000	N/A	0.052 / 0.000	N/A
Light	0.053 / 0.050	-5.3%	0.068 / 0.066	-2.1%
Aggressive	0.128 / 0.168	**+23.7%**	0.163 / 0.195	**+16.7%**

Fig. 3 shows this qualitatively: raw odometry visibly drifts from the straight ground-truth path, while the EKF estimate tracks it more closely.

Continuous error: the central finding. When error is instead measured at *every* odometry reading, the result reverses. Table 2 reports continuous-time error under aggressive drift, using the same recordings. Aggregated across all readings ($n = 1607$), the EKF is 12.7% *worse* than raw odometry by mean error and 13.4% worse by RMSE (Wilcoxon signed-rank, $p < 0.001$), the opposite sign from the correction-instant result, with RMSE confirming the same reversal rather than an artifact of the mean-error metric.

Table 2. Continuous position error (m), every odometry reading, aggressive drift. Each cell reports EKF / odom-only; RMSE change confirms the same reversal as mean error.

Scenario	**Cov.**	**Mean EKF/Odom**	**Mean Δ**	**RMSE EKF/Odom**	**RMSE Δ**
Stationary	27.6%	0.212 / 0.177	-19.3%	0.246 / 0.205	-19.8%
Straight	11.3%	0.212 / 0.178	-18.8%	0.245 / 0.206	-19.2%
Curve/occl.	0.0%	0.178 / 0.178	-0.0%	0.205 / 0.205	-0.0%
Aggregated	—	**0.200 / 0.178**	**$-$12.7%**	**0.233 / 0.205**	**$-$13.4%**

The error above follows the Absolute Trajectory Error (ATE) convention of Sturm et al. [15]: per-sample position error against ground truth, without a rigid alignment step, appropriate here because estimate, odometry, and ground truth already share a common world frame via the fixed camera (unlike SLAM, where the trajectory has a free gauge). ATE alone cannot distinguish whether degradation comes from error accumulating over time or from each step already drifting more. We therefore also report Relative Pose Error (RPE) [15]: the

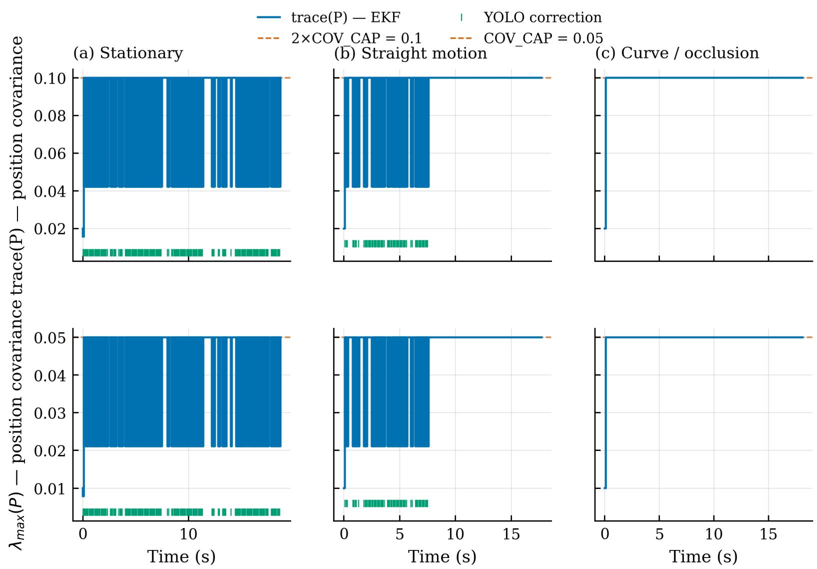

Fig. 4. Position-covariance saturation over time (step plots), all three scenarios, under two metrics (top: trace(P_{xy}); bottom: $\lambda_{max}(P_{xy})$, the largest eigenvalue), following Censi's [7] observation that the critical observation threshold depends on the chosen error metric. Short vertical ticks along the bottom of each panel mark successful YOLO corrections. Dashed lines mark each metric's ceiling under the same element-wise clip (Section 3): $2 \times \text{COV\_CAP} = 0.1$ for the trace (sum of both eigenvalues under near-isotropic P_{xy}) and $\text{COV\_CAP} = 0.05$ for the largest eigenvalue alone. Both metrics agree qualitatively: coverage decreases left to right (27.6% $\rightarrow$ 11.3% $\rightarrow$ 0%), and covariance saturates at its ceiling for correspondingly longer stretches, reaching permanent saturation in the zero-coverage case.

error of the position delta between consecutive readings, isolating local drift from the accumulated global error ATE captures. Aggregated across scenarios, RPE is 4.2% worse by mean error and 4.8% worse by RMSE, the same sign as ATE but substantially smaller, indicating the reversal is driven mostly by error accumulating over time under sparse correction, not by step-to-step drift.

Two observations explain this reversal. First, the degradation scales inversely with YOLO coverage: Scenario 3 (0% coverage) shows a change of exactly 0.0%, because with zero corrections the EKF's prediction degenerates to pure odometry-delta integration, mathematically identical to the unfused baseline. Scenarios 1 and 2 (27.6% and 11.3% coverage) show similar $\sim$19% degradation despite different coverage levels, suggesting the effect saturates quickly once corrections become sparse enough for the covariance ceiling (Section 3) to bind.

Second, and mechanistically explaining the first, Fig. 4 shows why: trace(P) saturates at the covariance ceiling almost immediately after any gap without correction, and only drops when a correction lands. In Scenario 3, the trace saturates within the first observed step and never drops again. This is precisely the divergence regime predicted by Sinopoli et al. [6] for observation arrival rates below a critical threshold: the covariance ceiling prevents unbounded growth (and the association failures it would otherwise cause) by capping the filter's *own estimate of its uncertainty*, not by making its state estimate more accurate during the uncorrected interval: the filter remains overconfident in its odometry-only prediction relative to how uncertain that prediction actually is.

Discussion. The natural reading of Table 2 is unflattering: our EKF, evaluated continuously, loses to the odometry it should improve upon. We argue this is expected, not a flawed design. Dissanayake et al.'s [5] convergence proof assumes dense observation; our camera violates that by construction (0–28% coverage), and Sinopoli et al. [6] predict exactly this covariance divergence without a ceiling, and resulting overconfidence with one. Scenario 3's 0.0% change is a clean confirmation: with zero corrections, the prediction step is provably identical to odometry-delta integration. We follow Censi [7] in reporting trace(P) as a scoping choice, not a claim of metric-independent divergence.

No standard alternative resolves the zero-coverage case: a particle filter degenerates to pure process propagation with no likelihood to reweight against; an IMM filter still requires some observation to weight competing modes; and a UKF targets observation-model nonlinearity, not observation *rate*, a distinction supported by the fact that an analogous critical-rate divergence has been shown for the UKF under intermittent observations [3]. We treat low coverage as the root cause, consistent with sensor-placement literature: estimation limits under a given sensor topology hold independently of the filter used. The covariance ceiling is thus an engineering approximation of maintaining a bounded, predictable state estimate during observation gaps, trading optimality for association stability, and inherits a known gating limitation (Altendorfer and Wirkert [9]): a saturated covariance widens the acceptance region precisely when true uncertainty is least well characterized by a fixed value, though we did not observe incorrect cross-robot association in our three scenarios.

We tested this trade-off directly rather than only arguing it. A fading factor (Adaptive Fading Kalman Filter), the textbook-correct alternative because it preserves the positive semi-definite structure of P, was implemented and evaluated under the same protocol. Without a ceiling, gain fell from +23.7%/+16.7% (clip) to −18.5%/−112.5%. A looser ceiling recovered a positive mean-error gain at the tightest setting tested (0.1: +9.1% mean but −38.8% RMSE, indicating occasional large outliers), then degraded again as the ceiling loosened further (0.2: −3.9%; 0.3: −19.3%/−111.3%). No configuration matched the clip on both metrics simultaneously, and the two metrics disagreed in sign at the one setting where mean error improved. This non-monotonic behavior is itself evidence against the fading factor as a drop-in replacement: its practical behavior depends sensitively on an additional hyperparameter with no principled default, and a looser ceiling still lets P grow enough for Mahalanobis-gated association to accept a neighboring robot's detection, the same failure mode the clip exists to prevent. We report this as empirical evidence, not only theoretical justification, for choosing the clip.

5 Future Work and Conclusion

Future work. Concrete directions include: (a) **adaptive process noise**: scale Q by time since the last correction instead of a fixed ceiling, more principled but, as our fading-factor test shows, not a guaranteed improvement without further tuning; (b) **LiDAR as a third fusion source**: each robot already publishes unused 2D scans; mutual LiDAR corrections (in the spirit of Khosravi et al. [10]) could target the zero-coverage regime directly; (c) **formal projection uncertainty**: propagate the full projection Jacobian into R; (d) **distributed relative localization**: Dobrynin and Garcia's [12] camera-free filtering on the same platform could provide a second correction channel during visual-coverage gaps.

Conclusion. We replaced a discrete visual/odometry fallback with a two-stage validated Extended Kalman Filter, evaluated on real recorded data from three TurtleBot3 robots. The filter delivers a genuine gain at correction instants (+23.7%) but is measurably worse than raw odometry when evaluated continuously (−12.7%) under 0–28% visual coverage, including a zero-coverage scenario collapsing, exactly as expected, to the unfused baseline. Rather than report only the favorable metric, we characterize both and ground the reversal in established theory on intermittent-observation Kalman filtering. EKF fusion remains the right tool for this problem, but its correctness under sparse observation must be measured and reported explicitly, and the observation rate itself, not the choice of filter, is the binding constraint future work should target.

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